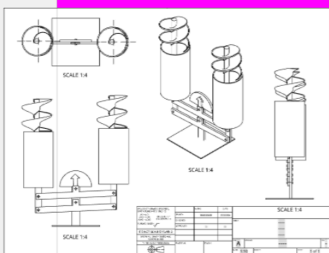
Introduction/Background: Dynamics. Uniqueness. Demonstration. These are the pillars of our project as to give soul to a machine, the machine must be dynamic, moving, and clearly demonstrate the winner at all times. Dynamic motion keep the middle schoolers engaged in being in the library and incentivizes them to visit, an important part of society due to the all time lows of kids reading. A machine that appears more responsive and immersive will increase library visits and thus improve literacy. In addition, the machine must clearly demonstrate the winner at all times so that results are immediate, understandable, and visually exciting for the audience. There have been previous solutions but most of them are boring with just dropping pom poms or just checkmarks, not really appealing to anyone. There needs to be something moving something interesting that automatically tabulates the results yet is mechanical aka a Roberval balance.

Problem: We were tasked with creating a voting machine for a middle school library.

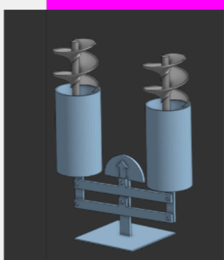


Needs to be stable.	0.25
Needs to be durable	0.2
Anything the kids have to access has to be 18 inches or shorter, 10 inches in length, 14 inches in width	0.25
Needs to hold ~300 pom poms	0.04
Needs to be simple for the kids to use	0.15
Needs to show which side is winning	0.01
Needs to be low-maintenance for the librarian	0.1

	Design Option A	Design Option B	Design Option C	Design Option D	Design Option E	Design Option F
total team scores	1.56	1.61	1.26	1.17	1.53	1.08

Testing:
We worried about the scale being fragile. It ended up being pretty durable and just needs some smoothening out for joints. 300 mini pom poms can fit in the quart sized mason jars easily.

Video Link
https://drive.google.com/file/d/1QEKCBu_f7Dh6U1jEajw6A-EByUeFrgqb/view?usp=sharing



Our 8 Initial Ideas:
1. A balance scale that tips to show which side has more pom poms.
2. Pom poms will slide into the jars down a slide similar to something found in a gumball machine.
3. The scale will have a perpendicular arrow attached to its center that will point at the sign of the winning choice.
4. An indicator card could be placed on the front of each jar on the scale containing the pom poms. As the Pom Poms fill the jar they will hide the indicator.
5. As either side of the scale goes down, it lifts up an indicator to show how much each vote is winning or losing.
6. Again using the weight idea we use a gear ratio to make an arrow point at the winning decision.
7. Have a pom pom launcher. Two launchers each with a button where you put the pom pom into the corresponding launcher and press the button.
8. A tree-like construct to hold up the signs. It'll be a pillar attached to the base of the scale and have two branches at a 120 degree angle. It's basically a glorified Y shape.

First CAD Model (click google drive Icon above)

First CAD Design

Orthographic Drawing

First Prototypes (oldest t)

The Final Version will be 3d Printed!!!